THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 12: **How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Deforestation leads to: 1

(a) loss of habitats and soil erosion **(b)** more oxygen
(c) more rainfall **(d)** cooler climates

2. Organisms that break down dead plants and animals are: 1

(a) decomposers **(b)** carnivores **(c)** producers **(d)** herbivores

3. Of the energy available at one trophic level, about how much passes to the next? 1

(a) 100 per cent **(b)** 50 per cent **(c)** 90 per cent **(d)** 10 per cent

4. Which of the following human activities harms an ecosystem? 1

(a) planting trees **(b)** dumping plastic waste in rivers
(c) composting **(d)** using solar energy

5. Which of the following is a decomposer? 1

(a) grass **(b)** fungi **(c)** deer **(d)** tiger

6. Food chains are usually short because: 1

(a) much energy is lost at each step **(b)** plants are few
(c) space is short **(d)** animals are lazy

7. Which is a way to conserve biodiversity? 1

(a) burning waste **(b)** cutting forests
(c) creating wildlife sanctuaries **(d)** hunting

8. The ultimate source of energy for almost all ecosystems is the: 1

(a) Moon **(b)** wind **(c)** soil **(d)** Sun

9. An organism that depends on another and harms it is a: 1

(a) herbivore **(b)** producer **(c)** decomposer **(d)** parasite

10. A habitat is: 1

(a) a kind of soil **(b)** a type of energy
(c) a food chain **(d)** the natural home of an organism

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is meant by ecological balance? 2

12. Differentiate between a food chain and a food web. 2

13. Why does energy decrease along a food chain? 2

14. Why should we not remove all the predators from an area? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the water cycle in brief. 3

16. Construct a food web using the organisms: grass, grasshopper, mouse, frog, snake and hawk. 3

17. What are the reasons for establishing national parks and wildlife sanctuaries? 3

OR

Explain how carbon dioxide is exchanged between plants and animals. 3

18. What happens to a grassland if all the frogs are removed? Explain. 3

OR

Differentiate between an aquatic and a terrestrial ecosystem with two examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the nitrogen cycle in brief and the role of bacteria, lightning and the farmer. 5

OR

Compare a pond ecosystem and a forest ecosystem on producers, consumers, decomposers and abiotic factors. 5

20. Describe the effects of human activities on ecosystems such as deforestation, pollution, overfishing and urbanisation. 5

OR

Explain the 3 R's and sustainable practices to restore the balance of nature. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND